Zoobenthic Community Indicators of Sediment Contamination

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Table 1.1: Table1 Pca Loadings

var	PC1	PC2	PC3	PC4	PC5	PC6
%OC	0.2398	0.0102	-0.0619	0.1267	0.0017	-0.0274
1234-TCB	0.0615	-0.0146	0.0493	0.2244	0.1144	0.7924
1245-TCB	0.1046	0.3572	0.0924	-0.2376	0.0223	-0.0311
Al	0.2492	-0.0293	-0.189	0.0399	0.1703	-0.1193
As	0.1424	-0.1519	-0.0708	-0.2066	0.0894	-0.0452
Bi	-0.0995	0.2882	-0.0307	0.3978	-0.0163	-0.1473
Ca	0.1489	0.241	-0.0924	0.2842	0.0188	0.0651
Cd	0.1345	0.1453	0.2956	0.2916	-0.2262	0.0348
Co	0.253	-0.028	-0.1276	-0.0369	0.1393	-0.0863
Cr	0.2551	-0.1152	0.0486	-0.0123	-0.1472	0.0105
Cu	0.229	-0.1514	0.0189	-0.0401	-0.2513	-0.0427
Fe	0.2551	-0.0512	-0.0914	-0.1007	0.1471	0.1437
HCB	0.0663	0.3687	0.1068	-0.2045	0.1815	0.1606
Heptachlor Epoxide	-0.0389	0.033	0.2035	0.1062	0.5047	0.1616
Hg	0.1189	0.1751	0.3562	0.0092	-0.0668	-0.1937
K	0.2391	0.0428	-0.1782	0.1384	0.174	-0.1253
Mg	0.1706	0.216	-0.1355	0.2701	0.0167	-0.0569
Mn	0.2498	0.018	-0.1643	0.1023	0.0545	-0.0125
Na	0.2224	0.1312	-0.1279	0.0126	-0.0682	-0.121
Ni	0.2736	-0.0502	-0.0128	-0.0007	-0.056	-0.0151
OCS	0.0626	0.3425	0.0228	-0.3233	-0.0102	-0.0603
Pb	0.2124	-0.0839	0.233	-0.0629	-0.2868	0.0853
QCB	0.0745	0.3684	0.0934	-0.2719	0.1858	0.0257
Sb	-0.1723	0.227	0.0154	0.3304	-0.0858	-0.1821
V	0.2438	-0.0704	-0.1666	0.037	0.1685	-0.0154
Zn	0.2515	-0.0491	0.0976	0.0636	-0.1976	0.1026
mirex	0.0345	-0.111	0.3583	0.1752	0.1934	-0.1004
p,p'-DDD	0.1153	-0.1551	0.3442	0.0049	0.1645	-0.2055
p,p'-DDE	0.0656	-0.2257	0.3157	0.086	0.3694	-0.2138
total PCB	0.1727	0.0612	0.3152	-0.0677	-0.2141	0.1419

Table 1.2: Table2 Rda Axes Summary

Axis	Eigenvalue	Expl. (%)	Cumul. (%)	F-stat	p-value
RDA1	0.0208	36.0749	36.0749	2.6938	0.296
RDA2	0.0132	22.88	58.9549	1.7085	0.876
RDA3	0.0088	15.1661	74.1211	1.1325	0.997
RDA4	0.0073	12.6117	86.7328	0.9418	1
RDA5	0.0044	7.6547	94.3875	0.5716	1
RDA6	0.0032	5.6125	100	0.4191	1

Table 1.3: Table3 Rda Predictor Terms

Variable	Inertia	F-stat	p-value	RDA1	RDA2
PC1	0.3345	1.0545	0.404	0.706	-0.0236
PC2	0.4795	1.5113	0.125	-0.5998	0.2667
PC3	0.3042	0.9589	0.477	0.0826	-0.5524
PC4	0.4403	1.3878	0.18	-0.0164	0.3057
PC5	0.451	1.4215	0.155	-0.4908	-0.2401
PC6	0.2607	0.8216	0.588	0.4716	-0.5281

Table 1.4: Table3 Rda Statistics

Statistic	Value
R^2	0.1758
Adjusted R^2	0.0346
Total Inertia	13.4728
Constrained Inertia	2.369
Unconstrained Inertia	11.1037
N Reference Sites	42

Table 1.5: Table0 Env Cluster Anova

Variable	Cluster 0	Cluster 1	F-stat	p-value
Measured Depth (m)	4.24 ± 3.24	3.37 ± 2.20	1.19	0.281
Velocity at bottom (m/sec)_Imputed	0.13 ± 0.14	0.12 ± 0.10	0.08	0.778
Water DO Bottom (mg/L)	9.20 ± 1.26	9.63 ± 0.46	2.29	0.137
Temperature (oC)	20.70 ± 2.25	20.11 ± 0.96	1.34	0.252
MPS (Phi)	0.93 ± 1.22	1.44 ± 0.66	3.15	0.082.
LOI (%)	1.88 ± 1.19	1.50 ± 1.05	1.43	0.237
Sample size (n)	30	22		

Table 1.6: Table1 Taxa Cluster Anova

Variable	Cluster 0	Cluster 1	F-stat	p-value
Oligochaeta	4.32 ± 1.38	5.54 ± 0.65	29.56	0.0000***
Chironomidae	3.70 ± 1.76	4.88 ± 1.05	25.25	0.0000***
Nematoda	2.63 ± 1.79	2.35 ± 1.55	0.37	0.543
Dreissena	2.66 ± 2.18	0.11 ± 0.35	36.19	0.0000***
Amphipoda	1.70 ± 1.57	0.01 ± 0.04	34.54	0.0000***
Sphaeriidae	1.00 ± 1.11	0.91 ± 1.45	0.05	0.817
Gastropoda	1.44 ± 1.63	0.11 ± 0.29	17.25	0.0001***
Caenis	0.88 ± 1.49	0.12 ± 0.28	3.66	0.061.
Hexagenia	0.56 ± 1.11	0.51 ± 0.92	0.05	0.831
Hydropsychidae	0.67 ± 1.38	0.33 ± 0.84	0.51	0.480
Acari	0.39 ± 0.70	0.67 ± 0.93	3.13	0.083.
Turbellaria	0.78 ± 1.10	0.08 ± 0.25	9.45	0.003**
Other Trichoptera	0.50 ± 1.09	0.37 ± 0.93	0.11	0.736
Hydrozoa	0.58 ± 1.24	0.08 ± 0.18	1.89	0.175
Hirudinea	0.21 ± 0.52	0.11 ± 0.51	0.66	0.419
Ceratopogonidae	0.06 ± 0.35	0.02 ± 0.12	0	0.960
Sample size (n)	30	22		

Table 1.7: Table1 Rda Axes Summary

Axis	Eigenvalue	Explained (%)	Cumulative (%)	F-statistic	p-value	Significance
RDA1	0.0344	52.9311	52.9311	6.2749	0.001	**
RDA2	0.0121	18.6191	71.5502	2.2072	0.551	ns
RDA3	0.0072	11.1062	82.6565	1.3166	0.986	ns
RDA4	0.0058	8.8746	91.5311	1.0521	0.998	ns
RDA5	0.0034	5.1669	96.698	0.6125	1	ns
RDA6	0.0021	3.302	100	0.3914	1	ns

Table 1.8: Table2 Rda Terms Summary

Environmental Variable	Delta Inertia	F-statistic	p-value	RDA1 Coef	RDA2 Coef
MPS (Phi)	0.8754	3.1356	0.003	-0.887	0.0014
Measured Depth (m)	0.5242	1.8774	0.028	0.6703	-0.35
Velocity at bottom	0.3706	1.3272	0.2	0.5773	0.0589
Temperature (oC)	0.31	1.1102	0.338	-0.2675	0.7438
Water DO Bottom (mg/L)	0.2511	0.8995	0.53	-0.2128	-0.4867
LOI (%)	0.2295	0.8218	0.611	0.231	0.508

Table 1.9: Table4 Lda Variable Importance

Environmental Variable	Delta Wilks' Lambda	F-statistic	p-value	LD1 Coef
MPS (Phi)	0.0894	4.8716	0.0324	0.8826
Temperature (oC)	0.0461	2.5148	0.1198	-0.6173
Velocity at bottom	0.0143	0.781	0.3815	0.344
Measured Depth (m)	0.0108	0.5913	0.4459	-0.2866
LOI (%)	0.0084	0.456	0.503	-0.2272
Water DO Bottom (mg/L)	0.0026	0.1424	0.7077	0.1347

Table 1.10: Table5 Lda Confusion Matrix

	Pred. Cluster 0	Pred. Cluster 1
True Cluster 0	24	6
True Cluster 1	8	14

Table 1.11: Table6 Lda Classification Report (Note: Overall Accuracy = 0.7308; Number of sites = 52.0)

	Precision	Recall	F1-Score	Support
Cluster 0	0.75	0.8	0.77	30
Cluster 1	0.7	0.64	0.67	22
Accuracy	–	–	0.73	52
Macro avg	0.72	0.72	0.72	52
Weighted avg	0.73	0.73	0.73	52

Table 1.12: Table7 Mccv Confusion Matrix (Test Size: 20.0% per iteration and Note: Combined results from 1,000 cross-validation iterations)

	Pred. Cluster 0	Pred. Cluster 1
True Cluster 0	4245	1755
True Cluster 1	2257	2743

Table 1.13: Table8 Mccv Classification Report (Note: Classification metrics shown as mean ± standard deviation across 1,000 CV iterations)

	Precision	Recall	F1-Score	Support
Cluster 0	0.661 ± 0.137	0.708 ± 0.202	0.669 ± 0.141	6
Cluster 1	0.638 ± 0.205	0.549 ± 0.215	0.564 ± 0.172	5
Weighted Avg	0.651 ± 0.151	0.635 ± 0.131	0.622 ± 0.137	11
Mean accuracy			0.635 ± 0.131	
Median accuracy			0.6364	

Table 1.14: Table1 Species Pc Loadings

Cluster	Taxon	PC1	PC2	PC3	PC4	PC5	PC6	PC7
0	Dreissena	0.558	0.5214	0.1035	-0.0131	-0.2544	-0.3619	-0.1815
0	Caenis	0.1021	-0.4493	0.2344	-0.1719	-0.1692	-0.2957	0.4786
0	Hexagenia	-0.1683	-0.0344	0.1718	-0.4159	0.5493	-0.3267	-0.1987
0	Gastropoda	0.2443	-0.2252	0.68	-0.0233	-0.0555	0.46	-0.1377
0	Nematoda	-0.3449	0.5399	0.2238	-0.2883	0.0076	0.0975	0.2117
0	Sphaeriidae	0.0226	-0.0604	0.2869	0.7051	0.2459	-0.354	0.0002
0	Amphipoda	0.3533	0.1152	-0.1715	0.0027	0.1557	-0.0829	0.5791
0	Other Trichoptera	0.1666	-0.208	-0.1909	-0.0855	0.1217	-0.183	-0.4273
0	Hydropsychidae	0.2933	-0.214	-0.3218	-0.2355	-0.0176	0.0447	-0.1525
0	Turbellaria	0.0374	0.1069	-0.1183	0.1717	0.5851	0.1372	0.118
0	Chironomidae	-0.2271	-0.197	-0.1273	-0.0756	-0.2119	-0.2629	0.1325
0	Oligochaeta	-0.3341	-0.0235	-0.161	0.2843	-0.2177	0.0139	-0.1551
0	Hydrozoa	0.25	-0.0298	-0.2017	0.0322	0.1725	0.4043	0.0641
0	Hirudinea	0.055	0.0446	0.1892	-0.1354	0.0418	-0.168	-0.1599
0	Acari	0.0352	-0.1392	0.0614	-0.1108	0.1744	-0.0289	0.0905
0	Ceratopogonidae	-0.0285	-0.0617	0.0653	-0.0785	0.0732	-0.0738	-0.014
1	Hexagenia	0.0318	0.3615	-0.3629	-0.5935	0.5124		
1	Nematoda	-0.649	0.1125	0.4104	0.0478	0.3939		
1	Sphaeriidae	0.6692	0.3492	0.4567	-0.029	0.1023		
1	Acari	-0.1957	0.5876	0.2206	0.1236	-0.4346		
1	Hydropsychidae	0.0295	0.1113	-0.3554	0.6241	0.2688		
1	Other Trichoptera	0.0764	0.3668	-0.4262	0.3087	-0.002		
1	Dreissena	-0.0689	0.1579	-0.1118	-0.1986	-0.367		
1	Caenis	-0.1821	0.0161	-0.2462	-0.1038	-0.245		
1	Oligochaeta	0.1023	-0.3702	-0.0314	0.0376	-0.1142		
1	Gastropoda	-0.0283	0.0929	-0.162	-0.1604	-0.1661		
1	Chironomidae	0.0665	-0.2309	-0.0981	-0.1628	-0.0419		
1	Hydrozoa	-0.026	0.047	-0.1025	-0.1049	-0.2263		
1	Ceratopogonidae	-0.0878	0.0088	-0.0467	-0.1649	-0.1398		
1	Turbellaria	-0.087	0.0924	0.0684	-0.0459	0.0121		
1	Hirudinea	0.124	-0.0738	0.0844	0.0029	0.01		
1	Amphipoda	0.0076	0.0452	-0.0143	-0.0214	-0.0505		

Table 1.15: Table2 Species Pc Pollution Regression

Cluster	PC	Var.(%)	R^2	$\bar{R}^2$	Coef.	t-stat	n
0	PC1	23.6	0.0751	0.0634	-0.0314	-2.53	81
0	PC2	13.6	0.0667	0.0549	-0.0225	-2.38	81
0	PC3	10.5	0.059	0.0471	-0.0186	-2.23	81
0	PC4	8.2	0	-0.0127	0.0002	0.02	81
0	PC5	7	0.0287	0.0164	-0.0106	-1.53	81
0	PC6	6.2	0.0005	-0.0122	0.0013	0.19	81
0	PC7	6.1	0.0186	0.0062	-0.008	-1.22	81
1	PC1	23.7	0.1062	0.0636	-0.0638	-1.58	23
1	PC2	17.8	0.0054	-0.042	0.0124	0.34	23
1	PC3	15.4	0.1456	0.105	-0.0603	-1.89	23
1	PC4	12.3	0.0196	-0.0271	-0.0197	-0.65	23
1	PC5	7.7	0.0062	-0.0411	-0.0088	-0.36	23

Table 1.16: Table 3a: Pollution PC vs Species PC Regression (Cluster 0)

Cluster	Poll. PC	Spec. PC	Var.(%)	R^2	Slope	n
0	PC1	PC1	23.6	0.0999	-0.0843	81
0	PC1	PC2	13.6	0.009	0.0192	81
0	PC1	PC3	10.5	0.001	-0.0057	81
0	PC1	PC4	8.2	0.0396	-0.0312	81
0	PC1	PC5	7	0.006	-0.0113	81
0	PC1	PC6	6.2	0.0076	-0.0119	81
0	PC1	PC7	6.1	0.024	0.0211	81
0	PC2	PC1	23.6	0.0139	-0.0288	81
0	PC2	PC2	13.6	0.0818	-0.053	81
0	PC2	PC3	10.5	0.0195	0.0227	81
0	PC2	PC4	8.2	0.0271	-0.0236	81
0	PC2	PC5	7	0.0028	-0.0071	81
0	PC2	PC6	6.2	0.0075	-0.0108	81
0	PC2	PC7	6.1	0.0375	-0.024	81
0	PC3	PC1	23.6	0.0407	-0.0488	81
0	PC3	PC2	13.6	0.0038	-0.0114	81
0	PC3	PC3	10.5	0.0774	-0.045	81
0	PC3	PC4	8.2	0.0438	0.0298	81
0	PC3	PC5	7	0.009	-0.0125	81
0	PC3	PC6	6.2	0.0017	0.0051	81
0	PC3	PC7	6.1	0.0026	-0.0063	81
0	PC4	PC1	23.6	0.0029	0.0138	81
0	PC4	PC2	13.6	0.0017	-0.008	81
0	PC4	PC3	10.5	0	0.0004	81
0	PC4	PC4	8.2	0.0057	-0.0114	81
0	PC4	PC5	7	0.0078	-0.0124	81
0	PC4	PC6	6.2	0.0012	-0.0045	81
0	PC4	PC7	6.1	0.0026	-0.0067	81
0	PC5	PC1	23.6	0.0034	-0.014	81
0	PC5	PC2	13.6	0.0518	-0.0417	81
0	PC5	PC3	10.5	0.0072	-0.0136	81
0	PC5	PC4	8.2	0.016	0.018	81
0	PC5	PC5	7	0.0056	-0.0098	81
0	PC5	PC6	6.2	0.0007	-0.0033	81
0	PC5	PC7	6.1	0.0036	-0.0074	81
0	PC6	PC1	23.6	0.008	0.0224	81
0	PC6	PC2	13.6	0.0034	-0.0112	81
0	PC6	PC3	10.5	0.0722	-0.045	81
0	PC6	PC4	8.2	0.0002	0.002	81
0	PC6	PC5	7	0.0045	-0.0091	81
0	PC6	PC6	6.2	0.0414	0.0262	81
0	PC6	PC7	6.1	0.0166	-0.0164	81

Table 1.17: Table 3b: Pollution PC vs Species PC Regression (Cluster 1)

Cluster	Poll. PC	Spec. PC	Var.(%)	R^2	Slope	n
1	PC1	PC1	23.7	0.0484	-0.0607	23
1	PC1	PC2	17.8	0	-0.0007	23
1	PC1	PC3	15.4	0.0332	-0.0406	23
1	PC1	PC4	12.3	0.0002	-0.0026	23
1	PC1	PC5	7.7	0.1592	0.063	23
1	PC2	PC1	23.7	0.0849	-0.1146	23
1	PC2	PC2	17.8	0.0077	0.03	23
1	PC2	PC3	15.4	0.1271	-0.1132	23
1	PC2	PC4	12.3	0.0631	-0.0711	23
1	PC2	PC5	7.7	0.0526	-0.0516	23
1	PC3	PC1	23.7	0.0593	0.1086	23
1	PC3	PC2	17.8	0.0004	0.0078	23
1	PC3	PC3	15.4	0.0043	-0.0235	23
1	PC3	PC4	12.3	0.0211	-0.0466	23
1	PC3	PC5	7.7	0.251	-0.1279	23
1	PC4	PC1	23.7	0.0608	-0.0607	23
1	PC4	PC2	17.8	0.08	0.0603	23
1	PC4	PC3	15.4	0.3287	-0.1139	23
1	PC4	PC4	12.3	0.0661	-0.0455	23
1	PC4	PC5	7.7	0	-0.0003	23
1	PC5	PC1	23.7	0.1568	-0.1837	23
1	PC5	PC2	17.8	0.0025	-0.0203	23
1	PC5	PC3	15.4	0.0402	0.0751	23
1	PC5	PC4	12.3	0.0238	0.0515	23
1	PC5	PC5	7.7	0.0482	-0.0583	23
1	PC6	PC1	23.7	0.0028	0.0157	23
1	PC6	PC2	17.8	0.0044	0.017	23
1	PC6	PC3	15.4	0.0391	-0.0471	23
1	PC6	PC4	12.3	0.0001	-0.0022	23
1	PC6	PC5	7.7	0.0096	0.0165	23

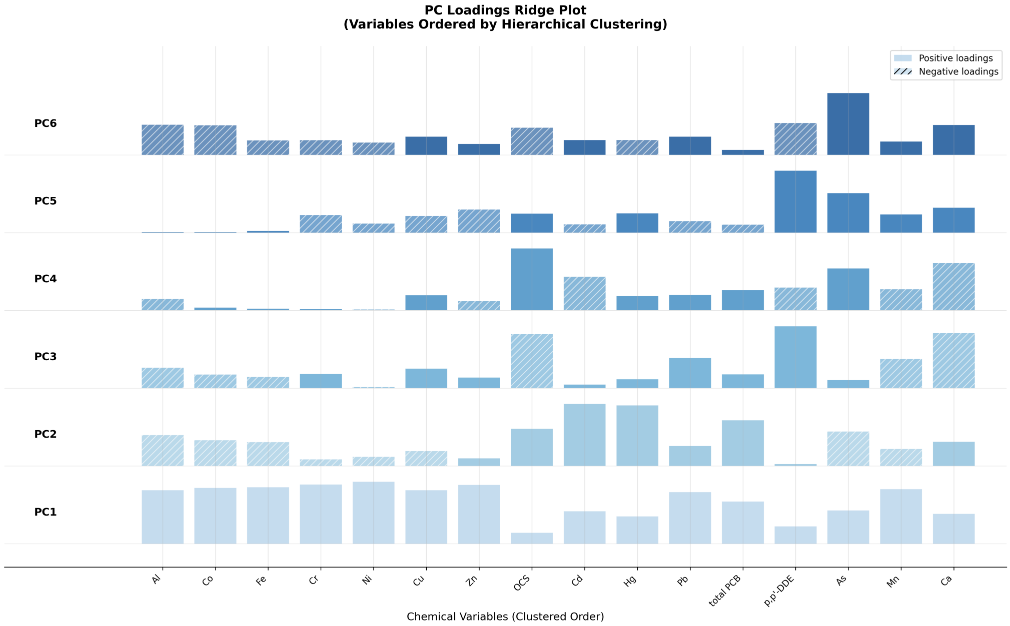


Figure 1.1: PC loadings ridge plot

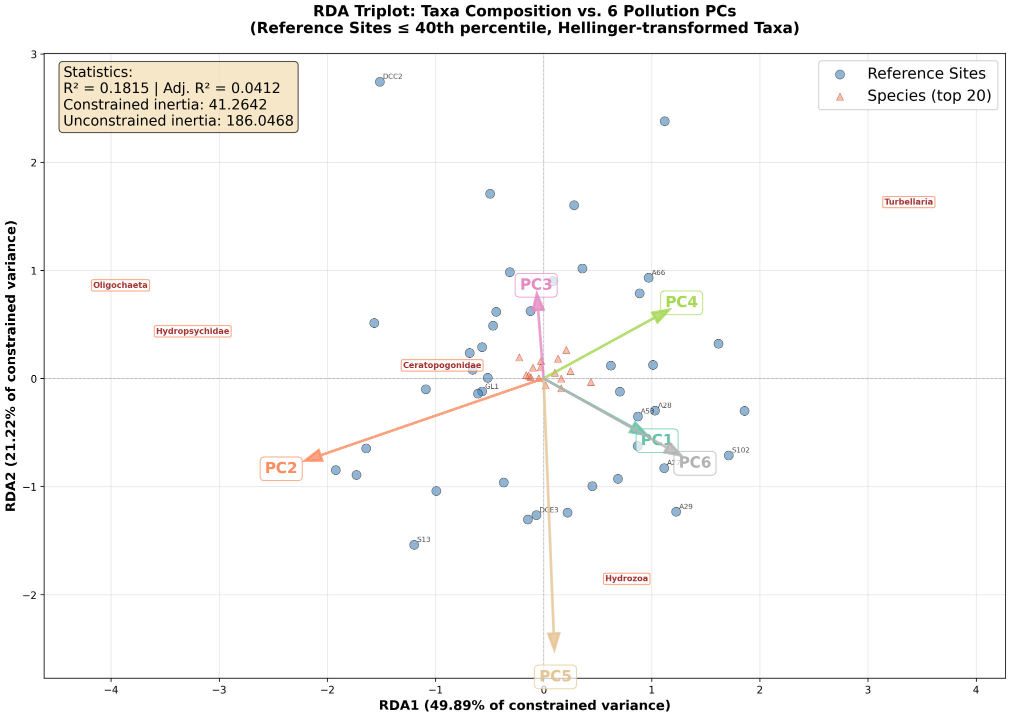


Figure 1.2: RDA validation plot of Pollution PCs vs Species Matrix

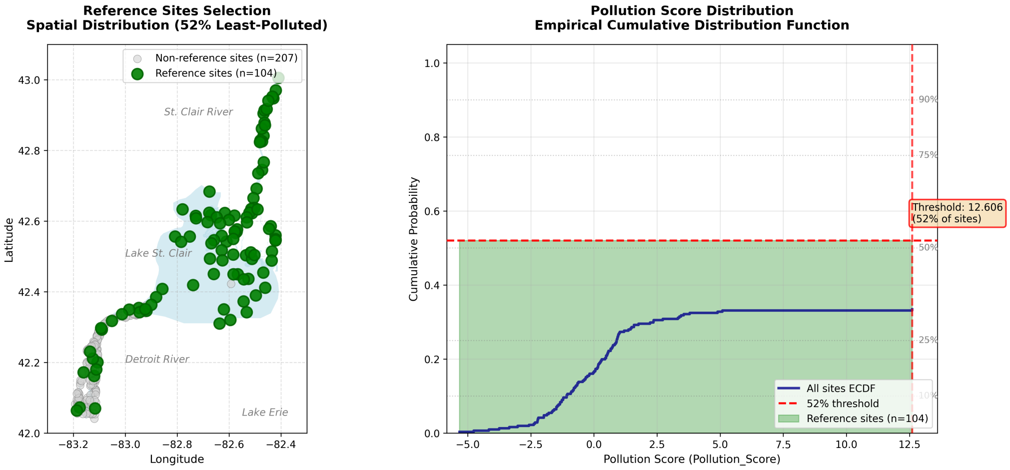
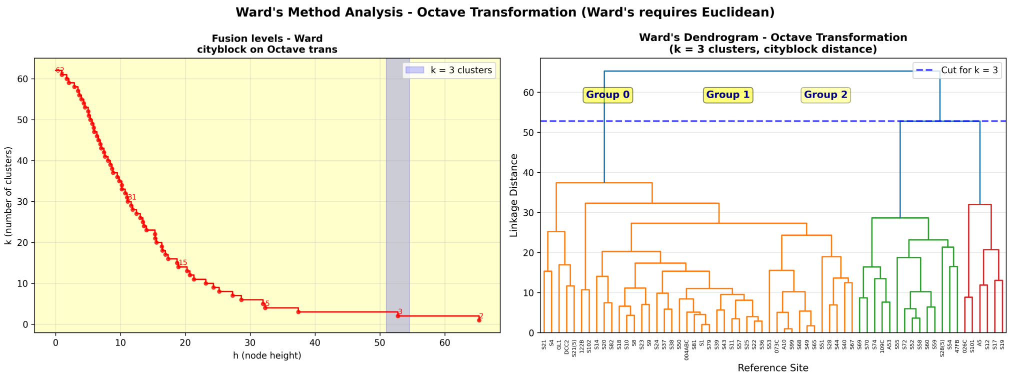
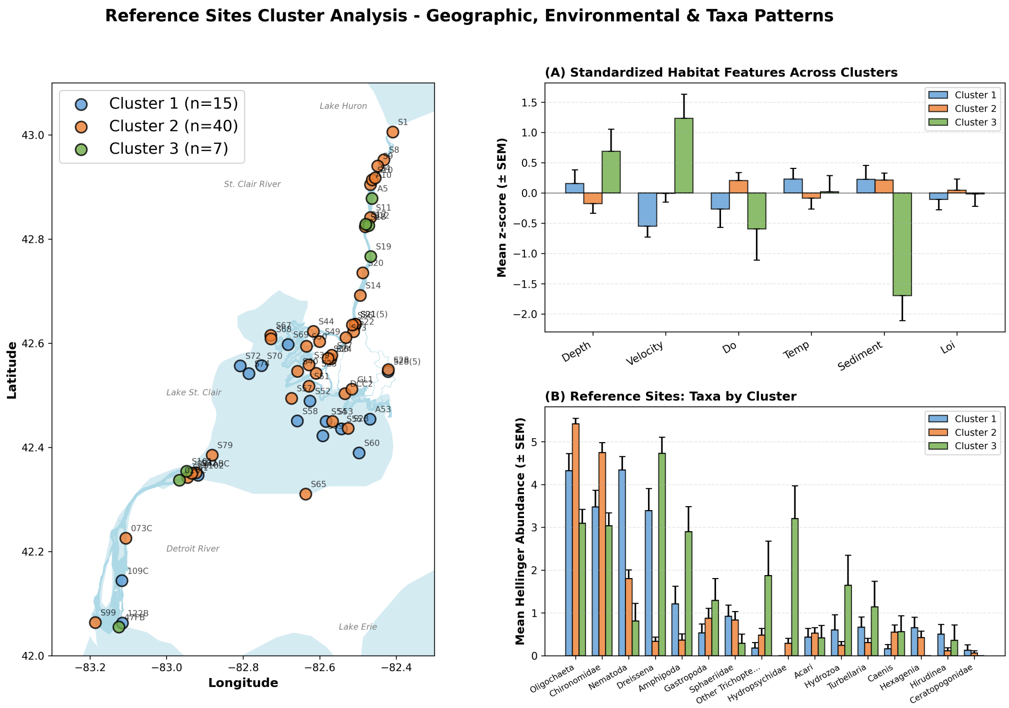


Figure 1.3: Reference site selection based on $p\%$ lowest pollution scores





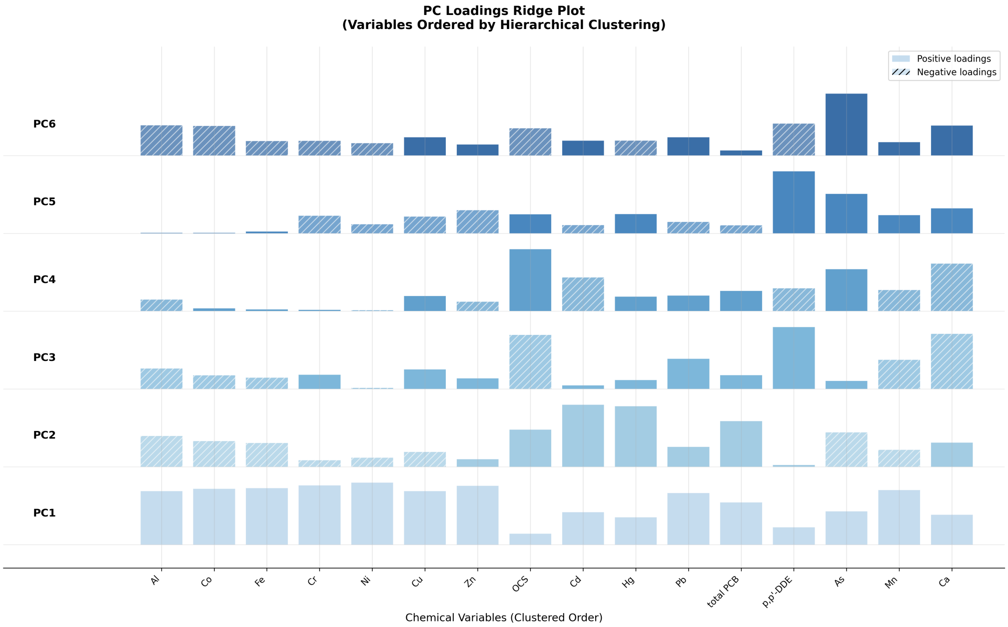


Figure 1.4: PC loadings ridge plot

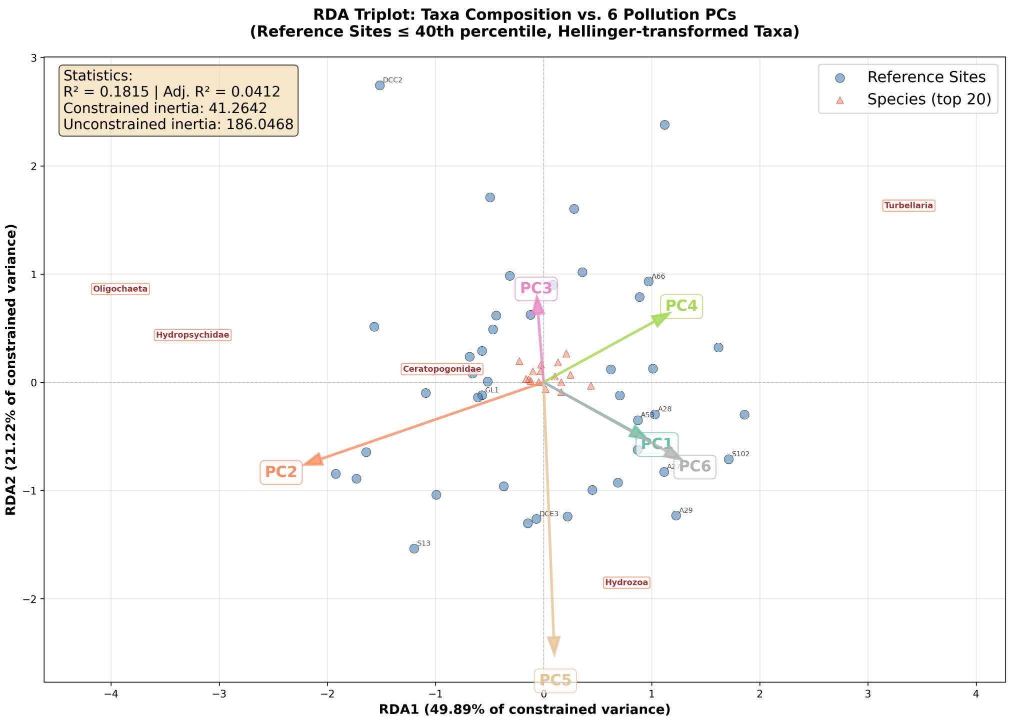


Figure 1.5: RDA validation plot of Pollution PCs vs Species Matrix

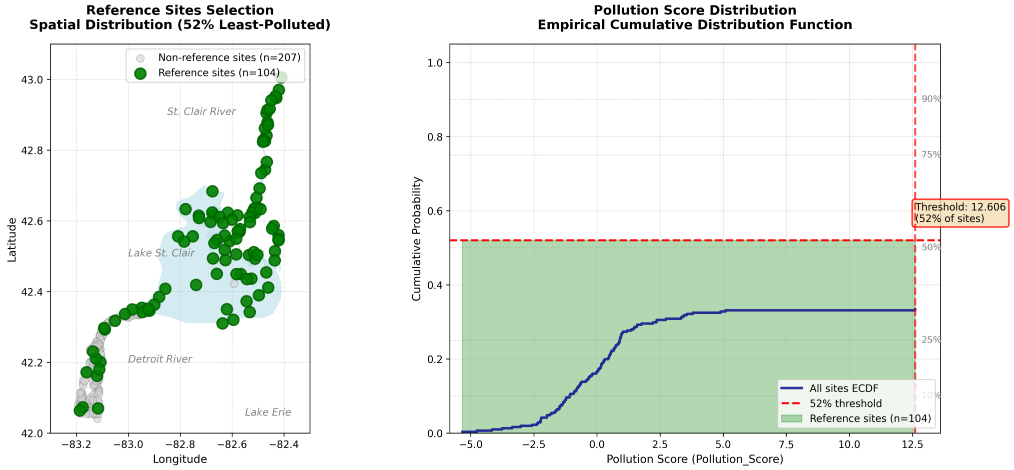
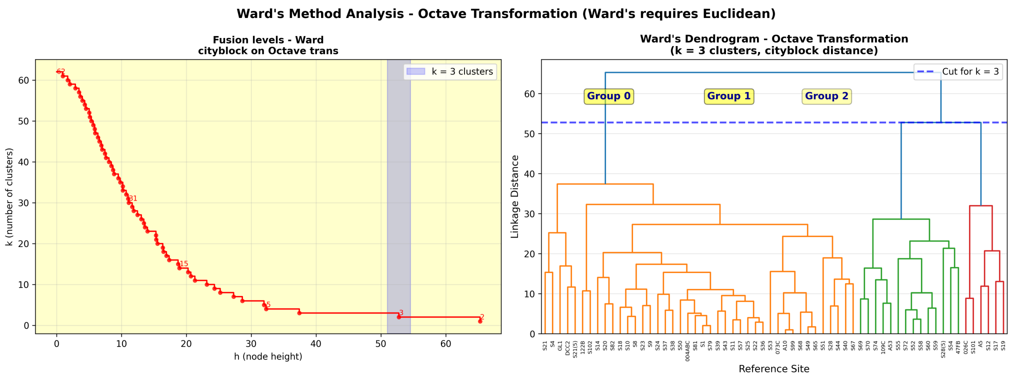
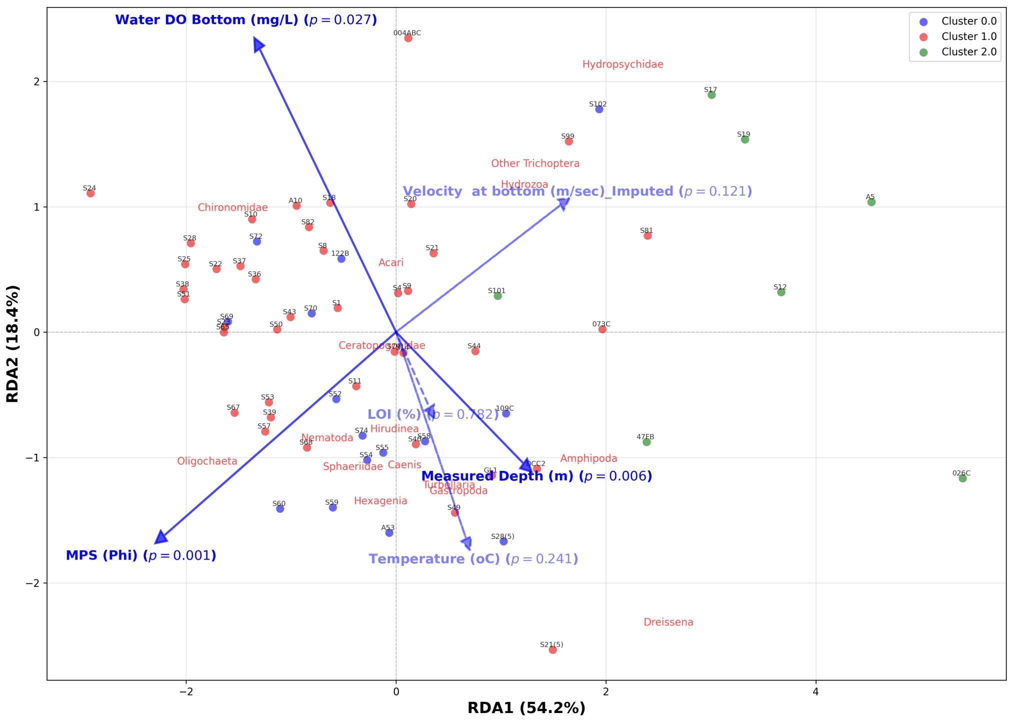
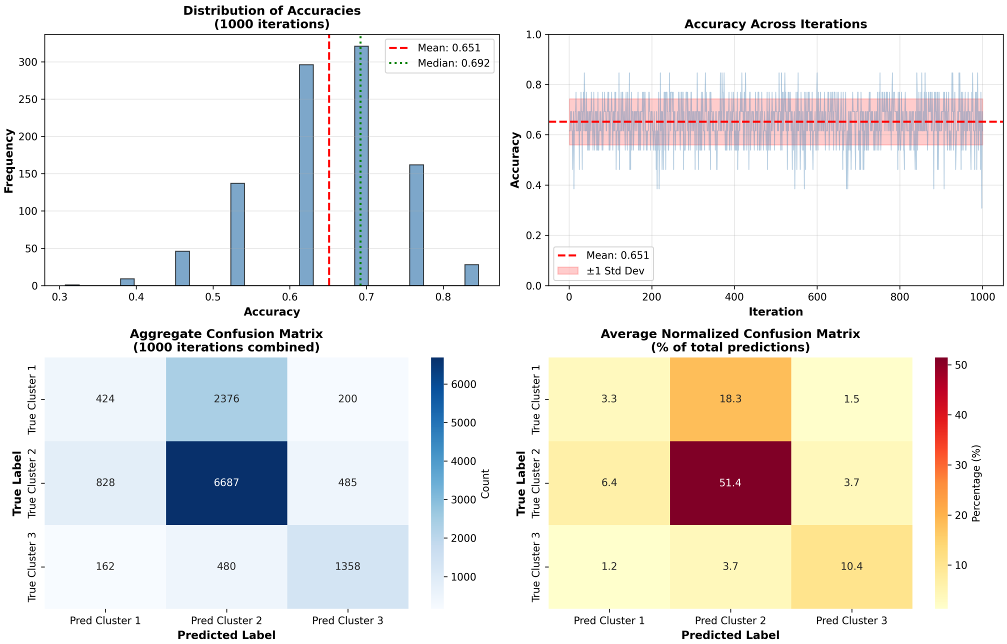


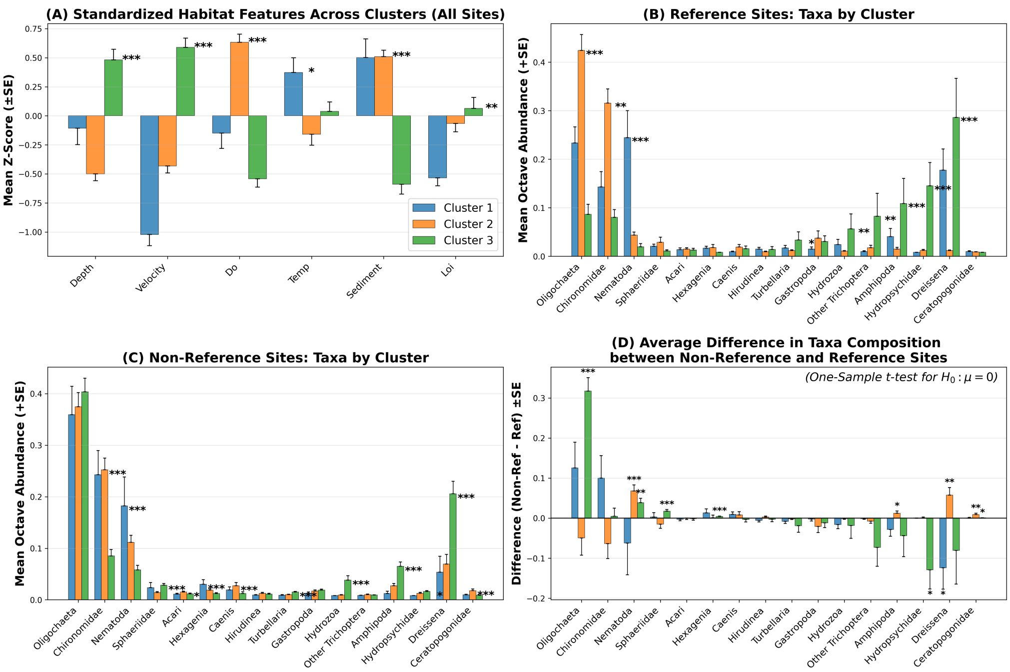
Figure 1.6: Reference site selection based on $p\%$ lowest pollution scores



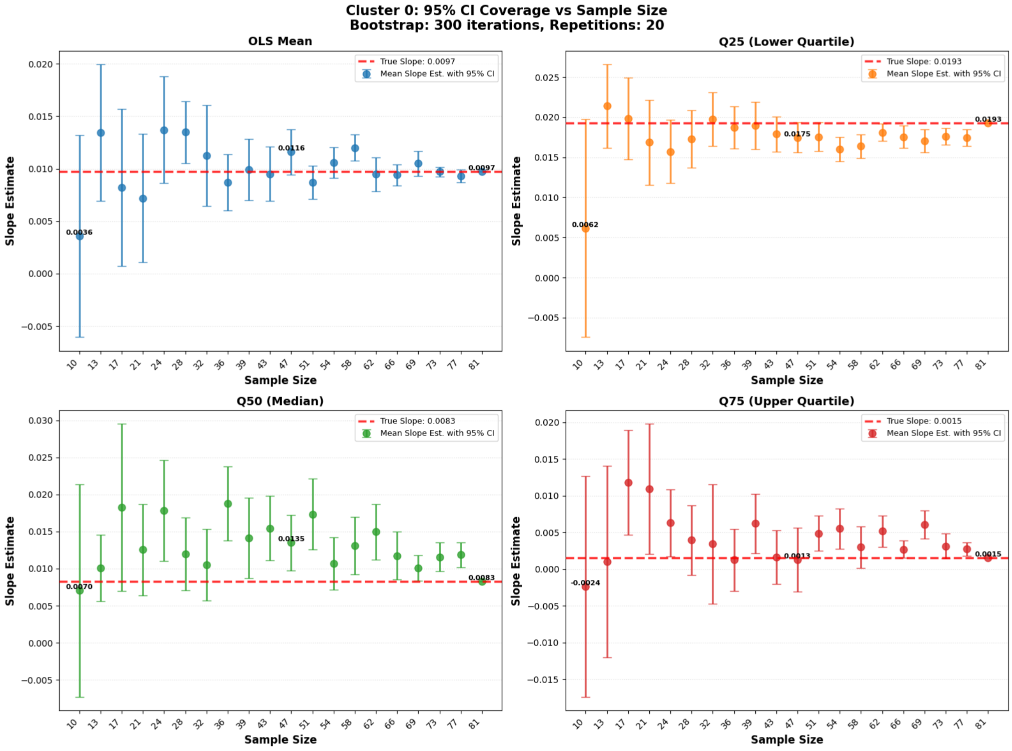


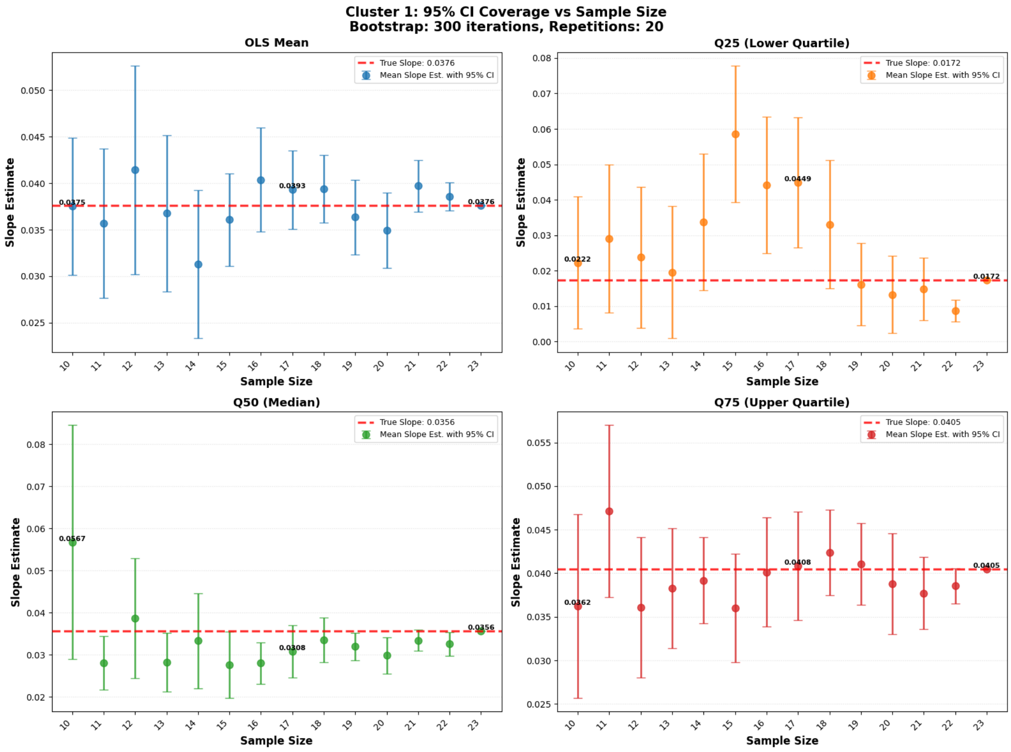


Distribution of Accuracies
(1000 iterations)
Mean: 0.651
Median: 0.692
Frequency
Accuracy
Accuracy Across Iterations
Accuracy
Iteration
Mean: 0.651
±1 Std Dev
Aggregate Confusion Matrix
(1000 iterations combined)
True Cluster 1
True Cluster 2
True Cluster 3
True Label
424
2376
200
828
6687
485
162
480
1358
Pred Cluster 1
Pred Cluster 2
Pred Cluster 3
Predicted Label
Count
Average Normalized Confusion Matrix
(% of total predictions)
3.3
18.3
1.5
6.4
51.4
3.7
1.2
3.7
10.4
Percentage (%)



(A) Standardized Habitat Features Across Clusters (All Sites)
Mean Z-Score (±SE)
Cluster 1
Cluster 2
Cluster 3
Depth
Velocity
Do
Temp
Sediment
Loi
(B) Reference Sites: Taxa by Cluster
Mean Octave Abundance (+SE)
Oligochaeta
Chironomidae
Nematoda
Sphaeriidae
Acari
Hexagenia
Caenis
Hirudinea
Turbellaria
Gastropoda
Hydrozoa
Other Trichoptera
Amphipoda
Hydropsychidae
Dreissena
Ceratopogonidae
(C) Non-Reference Sites: Taxa by Cluster
Mean Octave Abundance (+SE)
(D) Average Difference in Taxa Composition
between Non-Reference and Reference Sites
(One-Sample t-test for $H_0: \mu = 0$)
Difference (Non-Ref - Ref) ±SE


Cluster 0: 95% CI Coverage vs Sample Size
Bootstrap: 300 iterations, Repetitions: 20
OLS Mean
True Slope: 0.0097
Mean Slope Est. with 95% CI
Q25 (Lower Quartile)
True Slope: 0.0193
Q50 (Median)
True Slope: 0.0083
Q75 (Upper Quartile)
True Slope: 0.0015
Slope Estimate
Sample Size


Cluster 1: 95% CI Coverage vs Sample Size
Bootstrap: 300 iterations, Repetitions: 20
OLS Mean
True Slope: 0.0376
Mean Slope Est. with 95% CI
Q25 (Lower Quartile)
True Slope: 0.0172
Q50 (Median)
True Slope: 0.0356
Q75 (Upper Quartile)
True Slope: 0.0405
Slope Estimate
Sample Size